

TITAN-X

Stratospheric Persistent Aerial Platform

Master investor document — synthesis of the Consolidated Summary, regional deployment concept, and expanded 2026–2030 business plan. Planning estimates for qualified parties. Not an offer to sell securities.



Cover plate — California firefighting configuration (retardant + water delivery).

1. Executive summary

TITAN-X is a free-flight stratospheric aerostat operating in the altitude band between conventional aircraft and satellites. It carries a **20,000 lb** modular payload, remains aloft indefinitely on helium buoyancy and solar power, and reconfigures by exchanging gondolas and underbelly modules.

Investment thesis: fund pathfinder validation → first full-scale hull → California firefighting concentration + metro pre-positioning nodes → planning **12-unit** fleet by 2030. Pre-positioning converts response from days into hours. Civil and commercial utilization is intended to underwrite year-round availability. Dual-use configurations share one hull family and cost base.

\$13M Unit CAPEX (fleet volume)	\$156M 12-unit fleet CAPEX	\$2.4M Annual OPEX / unit
\$200K Monthly break-even / unit	17.8 : 1 Planning 10-yr fleet ROI	20,000 lb Net modular payload

Bottom line from source summary: Indefinite endurance. 20,000 lb modular payload. Near-zero operational carbon for persistent missions. Pre-positioned regional coverage with a California firefighting emphasis. Dual-use economics that allow civil and defense needs to share the same platforms.

2. Platform at a glance

Parameter	Value	Notes
Size	480 ft × 200 ft	Swept manta planform
Net payload	20,000 lbs	≈ school-bus weight
Altitude range	500 – 130,000 ft	Tree-top to lower stratosphere
Endurance	Indefinite	Solar + buoyancy
Ops carbon (persistent)	Near-zero	No combustion for loiter
Unit CAPEX (fleet volume)	\$13 million	12-unit fleet ≈ \$156M
Annual OPEX per unit	\$2.4 million	Break-even ~\$200K/month
Architecture	Fully modular	Gondola / underbelly module swap

2.1 Core advantages

- Indefinite endurance — solar power is sufficient for continuous station-keeping and operations
- High payload — 20,000 lbs supports sensors, supplies, cabins, or drone loads
- Fully modular — one hull family serves many missions by changing modules
- Near-zero operational carbon for persistent missions
- Rockoon capability — loft rockets to ~100,000 ft for major propellant savings, then recover the platform
- Dual-use architecture — civil, commercial, and defense requirements share the same platform and cost base

2.2 How the system works (investor framing)

Lift comes from helium buoyancy; electric/solar energy covers station-keeping, sensors, and module systems for persistent loiter without combustion. Mission packages attach as gondolas or underbelly modules — firefighting tanks and sensors one week, communications and observation the next — without building separate fleets. Operating altitude spans tree-top to lower stratosphere (500–130,000 ft), placing TITAN-X between aircraft sortie economics and satellite persistence.

3. Problem and market context

Wildfire and disaster response still depends on scarce surge aircraft and ground logistics. Assets often arrive late; coverage collapses when weather, tanker availability, or basing limits intervene. Satellites provide persistence but cannot deliver mass (retardant, water, logistics). Conventional aircraft deliver mass but not indefinite station-keeping. Cities and critical infrastructure need **hours-scale** airborne nodes for observation, temporary communications, and logistics — not ad-hoc charter after the event.

Gap: nothing practical today combines indefinite loiter, school-bus-class modular payload, rapid mission swap, and near-zero persistent ops carbon in the aircraft–satellite band. TITAN-X is positioned as persistent aerial **infrastructure**, not a one-off surge novelty.

3.1 Customer / payer segments

Segment	Primary use	Why TITAN-X
State / regional wildfire authorities	CA firefighting concentration	Sensors + retardant/water + incident command on station
Metro / high-value regions	Disaster & communications nodes	Hours-scale response vs days
Critical infrastructure operators	Pipelines, grids, ports, campuses	Continuous observation + logistics
Maritime / shipping	Domain awareness, ship support	Persistent coverage without ship sortie cost
Public safety	Low-cost drone threat awareness	Wide-area airborne node
Science / weather	High-altitude observation	Long dwell, recoverable platform
Rockoon / launch customers	Stratospheric loft	Propellant savings + recovery
Dual-use / defense (selected)	ISR, relay, forward logistics	Shared civil cost base

4. Regional deployment concept (source lock)

The preferred operational model is **pre-positioned regional coverage** rather than centralized surge response. Pre-positioning converts response time from days into hours. Civil and commercial utilization helps cover the cost of keeping assets available year-round. The same modular hull can shift between firefighting, disaster, infrastructure, and other configurations without separate fleets.

4.1 Major cities and high-value regions

Each major metropolitan area or high-risk region is assigned one TITAN-X (or a small pair). Units remain loaded with modular response packages and can be on station in hours. They function as persistent nodes for disaster response, communications restoration, wide-area observation, and logistics support.

4.2 California firefighting concentration

California is a natural early focus because of the scale and frequency of wildfire risk. A dedicated set of units would be configured primarily for firefighting support:

- Fire-behavior, thermal, and smoke-plume sensors
- Retardant or water delivery modules
- Incident command and communications packages
- Decision-support tools for incident commanders

These units remain modular and can shift to other missions in lower-risk periods, preserving utilization.

4.3 Smaller complementary platforms

Smaller aerostats or derivative platforms cover secondary roles that do not require the full 20,000 lb capacity: local observation, communications relay, weather and scientific sensing, and lower-cost regional coverage. Tiered system — full-size TITAN-X as primary regional nodes; smaller platforms for specialized or lower-intensity tasks.



Formation / persistent high-altitude operations — regional pre-positioning on one modular hull family.

5. Mission areas and configurations

- Disaster response and wildfire support
- Critical infrastructure monitoring
- Emergency and temporary communications
- Maritime domain awareness and ship support
- Public-safety airborne threat awareness (especially low-cost drones)
- Scientific and high-altitude observation
- Rockoon launch services
- Selected dual-use defense configurations (persistent ISR, communications relay, forward logistics, responsive access)

5.1 Firefighting CONOPS (planning)

Pre-positioned units over or near high-risk California regions provide continuous thermal/smoke awareness before ignition peaks, then deliver retardant or water modules under incident command. Decision-support tools feed commanders without requiring a separate ISR fleet. Off-season, the same hull swaps to metro disaster, infrastructure, or communications packages — preserving utilization and avoiding a single-mission stranded asset.

5.2 Rockoon and dual-use

Rockoon loft to ~100,000 ft targets propellant savings for responsive access, with platform recovery after launch. Dual-use defense options (ISR, relay, forward logistics) are selected configurations on the civil hull — not a separate product line — so civil utilization can help carry the cost base.

6. Development path and capital staging

Capital is staged: **pathfinder validation** → **first full-scale hull** → **initial operational units** → **fleet scale-up**. Civil and commercial utilization is intended to help underwrite the cost of maintaining persistent regional coverage.

Phase	Years	Milestone / exit
A. Pathfinder	2026	Endurance, station-keeping, modular swap demonstrated
B. First full-scale hull	2026–2027	Operational unit class
C. Initial nodes	2027–2028	CA firefighting layer + first metro nodes
D. Fleet scale-up	2028–2030	Toward planning 12-unit fleet + complementary platforms

Gate heavy fleet CAPEX on: endurance demo, modular swap demo, and first paid availability or mission contract.

7. Unit economics

Item	Planning value
Unit CAPEX (fleet volume)	\$13.0M
12-unit fleet CAPEX	\$156M
Annual OPEX per unit	\$2.4M
Break-even utilization	~\$200K / month (covers OPEX)
Mature utilization target (base)	\$8–12M / unit / year
Contribution at \$10M rev / \$2.4M opex	~\$7.6M / unit / year

7.1 Illustrative mature revenue mix

Stream	Share of unit revenue	Role
Regional standby / availability	35–45%	Underwrites year-round presence
Wildfire / disaster mission fees	20–30%	High intensity, seasonal

Infrastructure / maritime / public safety	15–25%	Steady civil utilization
Rockoon / science / specialty	5–15%	High-margin spikes
Dual-use / defense options	0–20%	Selected configs; shared hull

8. Financial plan 2026–2030

USD millions. Anchored to consolidated summary CAPEX / OPEX / break-even. The planning **17.8 : 1** ten-year fleet ROI is a long-horizon target on the stood-up fleet — not a 2030 cash guarantee. Early hulls cost more than the \$13M volume target; the model includes learning-curve premium.

8.1 Fleet build (base)

	2026	2027	2028	2029	2030
Full-size units EOY	0*	1	3	6	12
CA firefighting-primary	—	1	2	3	4
Metro / regional nodes	—	0	1	3	8

*Pathfinder year — endurance, station-keeping, modular swap validation.

8.2 Capital expenditure (base, \$M)

	2026	2027	2028	2029	2030	Total
Pathfinder + engineering	18	6	4	3	2	33
Hull production (learning curve)	—	18	30	42	78	168
Ground / ops infrastructure	4	6	8	10	12	40
Total CAPEX	22	30	42	55	92	241

8.3 Operating model (base, \$M)

	2026	2027	2028	2029	2030
Avg units in service	0	0.5	2.0	4.5	9.0
Revenue	0.5	4	18	42	90
Direct OPEX	1.5	3.5	7	13	24
Program / G&A	4	5	7	9	11
EBITDA (planning)	(5.0)	(4.5)	4	20	55

2030 check: ~\$90M revenue on ~9 average units ≈ **\$10M per unit-year**, above the \$2.4M OPEX break-even. Cumulative CAPEX 2026–2030 ≈ \$241M; cumulative revenue ≈ \$154.5M during build years; 2030 run-rate revenue ≈ \$90M.

8.4 Scenario fan — 2030

Case	Units	Revenue	EBITDA	Note
Conservative	6	\$45M	\$18M	Slower production; CA + few metros
Base	12	\$90M	\$55M	Matches summary fleet target
Upside	12+	\$130M	\$85M	Strong dual-use + rockoon + multi-state wildfire

9. Competition and differentiation

Alternative	Strength	TITAN-X edge
Tanker / scooping aircraft	Proven mass delivery	No indefinite loiter; weather & basing limits
Helicopters	Precision	Low payload-hours; expensive persistence
LEO / GEO satellites	Persistent eyes	No mass delivery; limited logistics
Tethered aerostats	Persistent local	Site-bound; lower payload / mobility
HAPS aircraft concepts	High altitude	Rarely combine 20k-lb modularity + rockoon + dual-use

Differentiation: indefinite solar/buoyancy endurance × 20,000 lb modularity × regional pre-positioning × shared civil–defense cost base.

10. Use of funds and milestones

Use	Share	Purpose
Pathfinder + first full-scale hull	35–45%	Technical de-risk
CA firefighting + first metro modules	20–30%	Revenue-bearing configurations
Production tooling / learning-curve	15–20%	Drive unit cost toward \$13M volume
Ops bases, helium, insurance, crew	10–15%	Year-round readiness
Corporate / BD / compliance	5–10%	Contracting with public & private buyers

11. Risks and guardrails

Risk	Guardrail
Airworthiness / airspace	Pathfinder before fleet claims
Helium & ground logistics	OPEX includes top-off; dual-sourcing plan
Early unit cost > \$13M	Learning-curve premium modeled; volume target explicit
Public contracting cycles	Civil commercial utilization underwrites standby
Over-claiming ROI	17.8:1 is planning 10-year fleet figure — not a 2026–2030 cash guarantee
Defense narrative crowding civil	Lead with regional civil utility; dual-use as selected configs

12. The ask

We are engaging venture capitalists and business owners to fund or co-develop:

- Pathfinder validation and first full-scale hull
- California firefighting concentration as the first revenue-bearing mission stack
- Metro pre-positioning nodes that turn TITAN-X into practical infrastructure

Return thesis: stand up the planning 12-unit fleet, clear ~\$200K/month per-unit break-even via availability + mission mix, and pursue the long-horizon 17.8 : 1 fleet ROI from persistent dual-use utilization — not from a single surge event.



High-altitude station-keeping — solar/buoyancy persistence in the aircraft–satellite band.

13. Document control — what this final merges

This FINAL plan is the master synthesis available in the Cursor/GitHub workspace. It merges: (1) TITAN-X Consolidated Summary · Regional Deployment Concept (August 2026 source tables and narrative), (2) expanded executive business plan with 2026–2030 financials and unit economics, (3) firefighting, formation, and stratosphere concept plates. If a richer offline “Version 3” PDF exists on another device, upload it to merge any remaining sections into this same master file — do not maintain parallel copies.

One file to send: TITAN-X-FINAL-Investor-Plan.pdf — Prime Field Technologies LLC · August 2026.